

Solutions (Exercise 4.)

(1)

$$(4.1) \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = \sum_{n=1}^{\infty} \frac{\frac{1}{2} \cdot 2^n}{3^n} = \frac{1}{2} \cdot \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{2} \cdot \frac{\frac{2}{3}}{1 - \frac{2}{3}},$$

because $|\frac{2}{3}| < 1$.

$$(4.2) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \quad \left. \vphantom{\sum_{n=1}^{\infty}} \right\} \text{ We show convergence by the definition!}$$

$$\begin{aligned} S_n &= \sum_{k=1}^n \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} = \left(1 - \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}\right) + \dots + \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right) \\ &= 1 - \frac{1}{\sqrt{n+1}} \rightarrow 1 \text{ as } n \rightarrow \infty \Rightarrow \text{The series is convergent and its sum is 1.} \end{aligned}$$

$$(4.3) \sum_{n=1}^{\infty} \frac{1}{n^2 + 3n} \quad \text{The general term is a ratio of two polynomials} \Rightarrow \text{Comparison Test!}$$

$$\frac{1}{n^2 + 3n} \leq \frac{1}{n^2}, \text{ where } \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \Rightarrow \text{the original series converges!}$$

$$(4.4) \sum_{n=1}^{\infty} \frac{(-2)^{n+1}}{(n+1)^n} \quad \left. \vphantom{\sum_{n=1}^{\infty}} \right\} \text{ Root Test!}$$

$$C_{\#} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{(-2)^{n+1}}{(n+1)^n} \right|} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{2} \cdot 2}{n+1} = 0 < 1 \Rightarrow$$

$\Rightarrow$ the original series is absolutely convergent.

$$(4.5) \sum_{n=2}^{\infty} \frac{1}{n \cdot \log(n)} \quad \text{Condensation Test!}$$

The new series:

$$\sum_{u=2}^{\infty} 2^u \cdot \frac{1}{2^u \cdot \log(2^u)} = \sum_{u=2}^{\infty} \frac{1}{\log(2) \cdot u} = \frac{1}{\log(2)} \cdot \sum_{u=2}^{\infty} \frac{1}{u},$$

which diverges $\Rightarrow$ The original series diverges!

$$(4.6.) \sum_{u=3}^{\infty} \frac{(-1)^u}{5^{u-2}} = \sum_{u=3}^{\infty} \frac{(-1)^u}{\frac{1}{25} \cdot 5^u} = 25 \cdot \sum_{u=3}^{\infty} \left(-\frac{1}{5}\right)^u =$$

$$= 25 \cdot \frac{\left(-\frac{1}{5}\right)^3}{1 - \left(-\frac{1}{5}\right)} = \dots \Rightarrow \text{it converges!}$$

$$(4.7.) \sum_{u=1}^{\infty} (-1)^u (\sqrt{u+1} - \sqrt{u}) \quad \left. \begin{array}{l} \text{Alternating series} \Rightarrow \\ \Rightarrow \text{Leibniz's } \cancel{\text{Rule!}} \\ \text{Test!} \end{array} \right\}$$

It is enough to show that this tends to zero!

$$\sqrt{u+1} - \sqrt{u} = \frac{u+1 - u}{\sqrt{u+1} + \sqrt{u}} = \frac{1}{\sqrt{u+1} + \sqrt{u}} \rightarrow 0 \text{ as } u \rightarrow \infty. \Rightarrow$$

$\Rightarrow$ The original series converges!

$$(4.8.) \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}} \Rightarrow \text{It is of the form } \sum_{n=1}^{\infty} \frac{1}{n^k},$$

where $k < 1 \Rightarrow$

$\Rightarrow$ the series diverges!

$$(4.9.) \sum_{u=1}^{\infty} \frac{3^{u-2}}{u!} = \frac{1}{9} \cdot \sum_{u=1}^{\infty} \frac{3^u}{u!} \quad \left. \begin{array}{l} \text{Ratio Rule!} \end{array} \right\}$$

$$d = \lim_{u \rightarrow \infty} \left| \frac{\frac{3^{u+1}}{(u+1)!}}{\frac{3^u}{u!}} \right| = \lim_{u \rightarrow \infty} \frac{3^{u+1}}{(u+1)!} \cdot \frac{u!}{3^u} = \lim_{u \rightarrow \infty} \frac{3}{u+1} = 0 < 1$$

$\Rightarrow$ the series converges!

(4.10) $\sum_{n=1}^{\infty} \frac{n^3+1}{4n^4+n^2}$ Comparison Test!

(2)

$$\frac{n^3+1}{4n^4+n^2} \geq \frac{n^3}{4n^4+n^2} \geq \frac{n^3}{4n^4+n^4} = \frac{1}{5} \cdot \frac{1}{n}$$

where $\sum_{n=1}^{\infty} \frac{1}{n} = +\infty \Rightarrow$ The original series diverges!

(4.11) $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ Comparison Test!

$$\frac{1}{n(n+2)} = \frac{1}{n^2+2n} \leq \frac{1}{n^2} \Rightarrow \text{The original series Converges!}$$

(4.12) $\sum_{n=1}^{\infty} \frac{2 \cdot (-3)^n}{4^{2n+1}} = \frac{1}{2} \cdot \sum_{n=1}^{\infty} \left(-\frac{3}{16}\right)^n$ where

$$|q| = \left| -\frac{3}{16} \right| < 1 \Rightarrow \text{The series converges!}$$

(4.13) $\sum_{n=1}^{\infty} \frac{\sqrt{2n^2+1}}{n+3} = \sum_{n=1}^{\infty} \underbrace{\frac{\sqrt{2+\frac{1}{n^2}}}{1+\frac{1}{n^2}}}_{\rightarrow \sqrt{2} \text{ as } n \rightarrow \infty}$

$\rightarrow \sqrt{2} \text{ as } n \rightarrow \infty \Rightarrow$
and $\sqrt{2} \neq 0$

$\Rightarrow$ The series diverges!

(4.14) $\sum_{n=1}^{\infty} \frac{n^2+4n+1}{2^{3n+2}} = \frac{1}{4} \sum_{n=1}^{\infty} \frac{n^2+4n+1}{8^n}$ } Root Test!

$$\sqrt[n]{\left| \frac{n^2+4n+1}{8^n} \right|} = \frac{\sqrt[n]{n^2+4n+1}}{8} \rightarrow \frac{1}{8} \text{ as } n \rightarrow \infty$$

and $\frac{1}{8} < 1 \Rightarrow$ The series Converges!

$$(4.15) \sum_{u=1}^{\infty} \frac{2^u u!}{u^u} \quad \text{Ratio Test!}$$

$$\begin{aligned} d &= \lim_{u \rightarrow \infty} \left| \frac{2^{u+1} (u+1)!}{(u+1)^{u+1}} \cdot \frac{u^u}{2^u \cdot u!} \right| = \lim_{u \rightarrow \infty} \frac{2 u^u}{(u+1)^u} = \lim_{u \rightarrow \infty} 2 \cdot \left(\frac{u}{u+1} \right)^u \\ &= 2 \cdot \lim_{u \rightarrow \infty} \left(1 + \frac{-1}{u+1} \right)^u = 2 \cdot \lim_{u \rightarrow \infty} \left[\left(1 + \frac{-1}{u+1} \right)^{u+1} \right]^{\frac{u}{u+1}} = \\ &= 2 \cdot \frac{1}{e} < 1 \Rightarrow \text{The series converges!} \end{aligned}$$

$$(4.16) \sum_{u=1}^{\infty} \binom{2u}{n} \frac{1}{2^u} \quad \text{Ratio Test!}$$

$$\begin{aligned} \left| \frac{\binom{2(u+1)}{u+1} \frac{1}{2^{u+1}}}{\binom{2u}{u} \frac{1}{2^u}} \right| &= \frac{\binom{2u+2}{u+1} \frac{1}{2^{u+1}}}{\binom{2u}{u} \frac{1}{2^u}} = \frac{(2u+2)(2u+1) \cdot (2u)!}{2^{u+1} \cdot (u+1)! (u+1)!} \cdot \frac{2^u u! u!}{(2u)!} = \\ &= \frac{(2u+2)(2u+1)}{2 \cdot (u+1)(u+1)} = \frac{2u+1}{u+1} \rightarrow 2 > 1 \Rightarrow \text{the series diverges!} \end{aligned}$$

$$(4.17) \sum_{u=0}^{\infty} \left(\frac{n-1}{u+1} \right)^{u^2} \quad \text{Root Test!}$$

$$\sqrt[u]{\left| \left(\frac{n-1}{u+1} \right)^{u^2} \right|} = \left(\frac{n-1}{u+1} \right)^u = \left[\left(1 + \frac{-2}{u+1} \right)^{u+1} \right]^{\frac{u}{u+1}} \rightarrow \frac{1}{e^2} < 1$$

$\Rightarrow$ The series converges!

$$\begin{aligned} (4.18) \sum_{u=1}^{\infty} \frac{n!}{n^n} \quad \text{Ratio Test:} \quad & \frac{(n+1)!}{(n+1)^{u+1}} \cdot \frac{n^n}{n!} = \left(\frac{n}{n+1} \right)^n = \\ &= \left[\left(1 + \frac{-1}{n+1} \right)^{u+1} \right]^{\frac{u}{u+1}} \rightarrow \frac{1}{e} < 1 \Rightarrow \text{The series converges.} \end{aligned}$$

$$(4.19) \sum_{n=1}^{\infty} \frac{(n+1)^2}{3n^4 + n} \text{ Comparison Test!}$$

(3.)

$$\frac{(n+1)^2}{3n^4 + n} \leq \frac{(n+n)^2}{3n^4} = \frac{4}{3} \cdot \frac{1}{n^2} \Rightarrow \text{The series converges!}$$

$$(4.20) \sum_{n=2}^{\infty} \frac{1}{n^2 \log(n)} \Rightarrow \text{Condensation Test!}$$

$$\sum_{n=2}^{\infty} 2^n \cdot \frac{1}{4^n \log(2^n)} = \sum_{n=2}^{\infty} \frac{1}{\log(2) \cdot 2^n \cdot n} = \frac{1}{\log(2)} \sum_{n=2}^{\infty} \frac{1}{2^n n}$$

Root Test =

$$\Rightarrow \sqrt[n]{\left| \frac{1}{2^n n} \right|} = \frac{1}{2 \cdot \sqrt[n]{n}} \rightarrow \frac{1}{2} < 1 \Rightarrow \text{The test series converges} \Rightarrow$$

$\Rightarrow$ the original series converges!

$$(4.21) \sum_{n=1}^{\infty} \frac{n^2 2^n - 6n!}{n^2 \cdot n!} = \sum_{n=1}^{\infty} \frac{2^n}{n!} - \frac{6}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{2^n}{n!} \text{ converges (Ratio Test!)} \quad \left. \begin{array}{l} \text{Use} \\ \text{Ratio Test!} \end{array} \right\}$$

$$\sum_{n=1}^{\infty} \frac{6}{n^2} \text{ converges (Comparison Test!)} \quad \left. \begin{array}{l} \text{Use} \\ \text{Comparison Test!} \end{array} \right\}$$

There's sum, that is, the original series converges!

$$(4.22) \sum_{n=1}^{\infty} \frac{n^{17}}{2^n + 3^n} \text{ Root Test!}$$

$$\sqrt[n]{\left| \frac{n^{17}}{2^n + 3^n} \right|} = \frac{(\sqrt[n]{n})^{17}}{\sqrt[n]{2^n + 3^n}} \rightarrow \frac{1}{3} < 1 \Rightarrow \text{The original series converges!}$$

(k.23) $\sum_{n=4}^{\infty} \frac{3^{n-3}}{(-2)^{2n}} = \frac{1}{27} \cdot \sum_{n=4}^{\infty} \left(-\frac{3}{4}\right)^n$, which converges, because of the inequality $|\frac{3}{4}| < 1$.

$$(k.24) \sum_{n=1}^{\infty} \log\left(\frac{n}{n+1}\right) = \sum_{n=1}^{\infty} \log(n) - \log(n+1)$$

By definition:

$$\begin{aligned} S_n &= \sum_{k=1}^n \log(k) - \log(k+1) = \log(1) - \log(2) + \\ &+ \log(2) - \log(3) + \dots + \log(n-1) - \log(n) = \\ &= \log(1) - \log(n) = -\log(n) \rightarrow -\infty \Rightarrow \\ &\text{as } n \rightarrow \infty \end{aligned}$$

$\Rightarrow$ the series diverges!

$$(k.25) \sum_{n=2}^{\infty} \frac{\log(1+\frac{1}{n})}{\log(n) \log(n+1)} = \sum_{n=2}^{\infty} \frac{\log(n+1) - \log(n)}{\log(n) \log(n+1)}$$

$$= \sum_{n=2}^{\infty} \frac{1}{\log(n)} - \frac{1}{\log(n+1)} \Rightarrow \text{definition!!}$$

$$S_n = \frac{1}{\log(2)} - \frac{1}{\log(3)} + \frac{1}{\log(3)} - \frac{1}{\log(4)} + \dots + \frac{1}{\log(n-1)} - \frac{1}{\log(n)}$$

$$= \frac{1}{\log(2)} - \frac{1}{\log(n)} \rightarrow \frac{1}{\log(2)} \text{ as } n \rightarrow \infty \Rightarrow$$

$\Rightarrow$ The series converges!

$$\begin{aligned}
 (4.26) \quad \sum_{n=1}^{\infty} \frac{2^n + 3^n}{6^n} &= \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n + \left(\frac{1}{2}\right)^n = \\
 &= \underbrace{\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n + \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n}_{\text{Both converge, because of}}
 \end{aligned}$$

Both converge, because of
 $\frac{1}{3} < 1$ and $\frac{1}{2} < 1$.

(4.)